Cloud LearnX

Mathematics

Circle Theorems and Similar solids — Practice Worksheet

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Topic: Circle Theorems and Similar solids

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

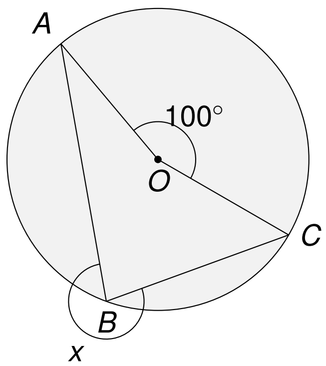
Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 *A*, *B* and *C* are points on the circumference of a circle with centre *O*.

AOC is the angle at the centre and *ABC* is the angle at the circumference, both standing on the same arc *AC*.



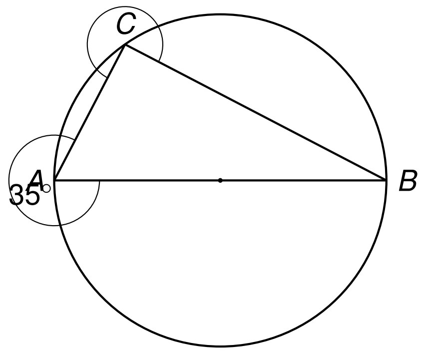
Find the size of angle x, giving a reason for your answer.

x = ... °

(2)

(Total for Question 1 is 2 marks)

2 An engineer is analysing the circular cross-section of a satellite dish support frame. Points A, B and C lie on the circumference of the circle, where AB is a diameter of the circle. A support strut runs along AC and another along CB.



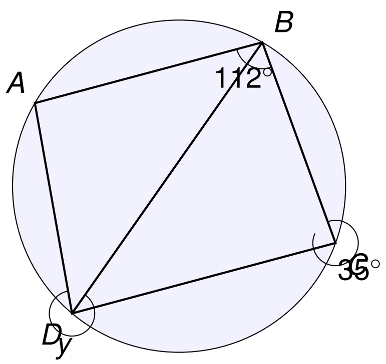
Given that angle $BAC = 35^\circ$, work out the size of angle ACB.

$ACB =$ .... °

(2)

(Total for Question 2 is 2 marks)

3 A, B, C and D are four points on the circumference of a circle, forming the cyclic quadrilateral $ABCD$.



Angle ABC = 112° and angle ACB = 35°.

(a) Work out the size of angle ADC.
Give a reason for your answer.

ADC = ... °

(2)

(b) Angle ADB (labelled y in the diagram) is subtended by the same chord as angle ACB.
Work out the size of angle y.
Give a reason for your answer.

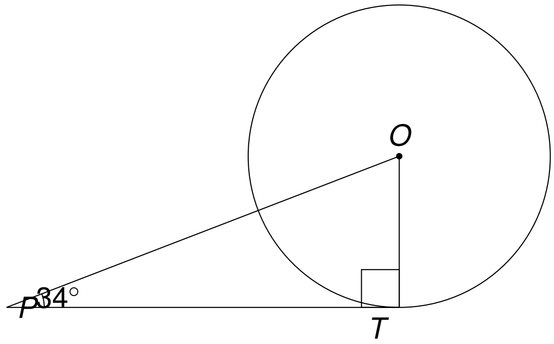
y = ... °

(2)

(Total for Question 3 is 4 marks)

4 A circular Valorant-themed drone light show display has its controller drone hovering at point T on the edge of a circular flight boundary, centre O. A signal tower is positioned at external point P, and the line PT is a tangent to the circular boundary at T.

The radius OT is drawn, and OP is drawn. The angle between the signal line PT and OP is 34°, as shown in the diagram.



(a) Write down the size of angle OTP, giving a reason for your answer.

OTP = ... °

(1)

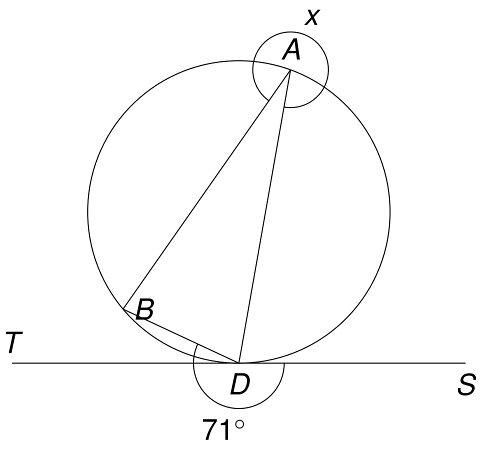
(b) Work out the size of angle TOP.

TOP = ... °

(2)

(Total for Question 4 is 3 marks)

5 A, B and D are points on the circumference of a circle. The line TDS is a tangent to the circle at D.
Chord DB and chord DA are drawn, and chord AB is drawn to complete triangle ABD.
The tangent-chord angle between TD and chord DB is 71°.



Show that x = 71°, giving reasons for each step of your working.

(3)

(Total for Question 5 is 3 marks)

6 An e-sports arena has two similar spherical trophy display cases, used to showcase the championship cups for a Valorant tournament. The two cases are mathematically similar in shape.

The small display case has a radius of 12 cm and contains a miniature trophy standing 9 cm tall inside it.

The large display case has a radius of 30 cm and contains a full-size trophy, geometrically similar in every way to the miniature one.

Work out the height of the trophy in the large display case.

.... cm

(3)

(Total for Question 6 is 3 marks)

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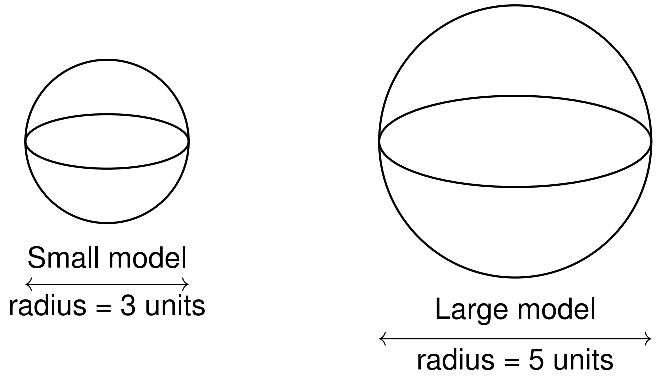
7 A science museum displays two similar spherical model planets, used to represent a planet and its larger counterpart in an exhibit about scale in the solar system. The two spheres are mathematically similar, with radii in the ratio 3 : 5.



The volume of the smaller model is 972 cm^3.

(a) Find the ratio of the surface area of the small model to the surface area of the large model.

....

(1)

(b) Find the ratio of the volume of the small model to the volume of the large model.

....

(1)

(c) Work out the volume of the larger model.

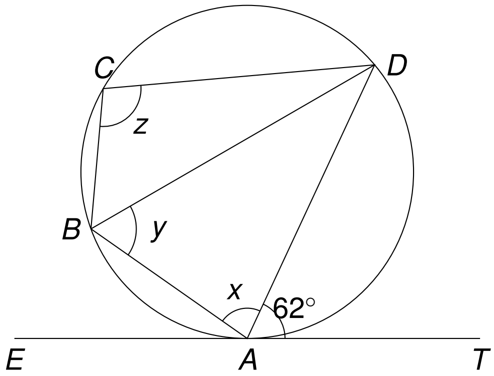
.... cm^3

(2)

(Total for Question 7 is 4 marks)

8 A satellite dish tracks a communication satellite that appears to trace a circular orbit path, as observed from a ground station. Points A, B, C and D lie on the circular path, and ET is a tangent to the circle at point A, representing the straight-line signal transmitted to the ground station at T.

$ABCD$ is a cyclic quadrilateral. The tangent ET touches the circle at A.



Angle $TAD = 62^\circ$ (the angle between the tangent AT and the chord AD).
Angle $ABD = 38^\circ$.

(a) Find the size of angle ADB (marked x is angle DAB; first find angle ADB), giving a reason for your answer.

ADB = ... °

(2)

(b) Find the size of angle DAB (marked x), giving a reason for your answer.

x = ... °

(1)

(c) Find the size of angle BCD (marked z), giving a reason for your answer.

z = ... °

(2)

(Total for Question 8 is 5 marks)

TOTAL FOR PAPER IS ?? MARKS